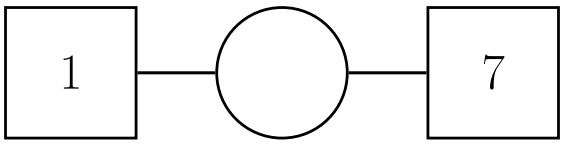
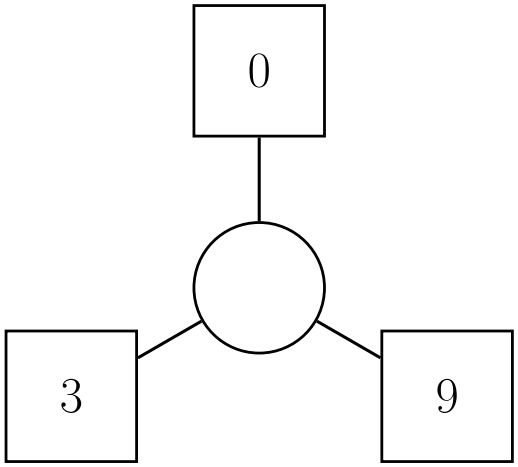
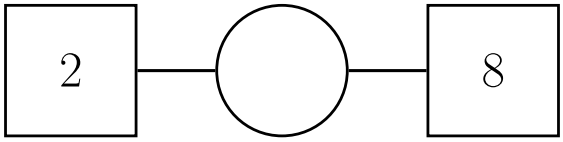
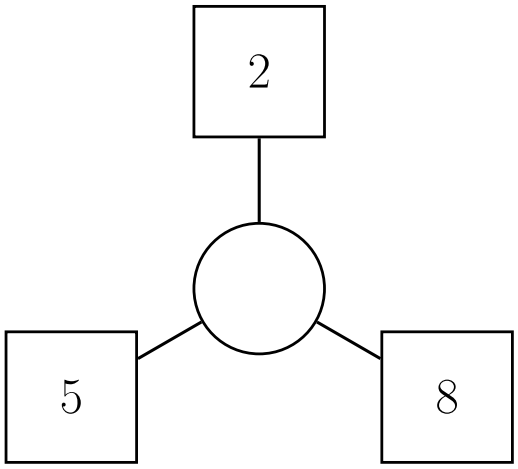
Add the numbers joined to a circle and divide by how many numbers you added. The answer must equal the circle's number. Squares keep their numbers. Every circle must work at the same time.

Problem 1: Fill the circles. Then change one square number on each board so that its circle number grows by 2.

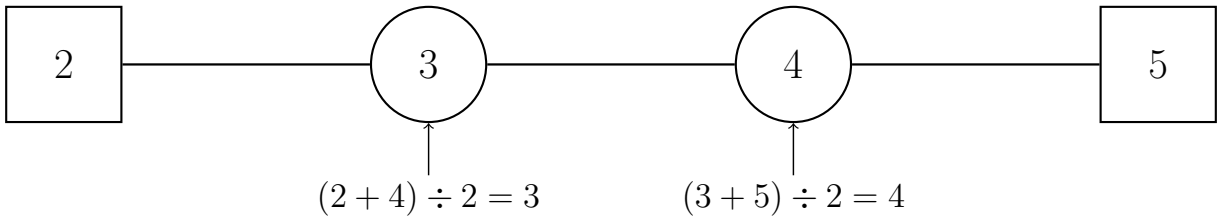




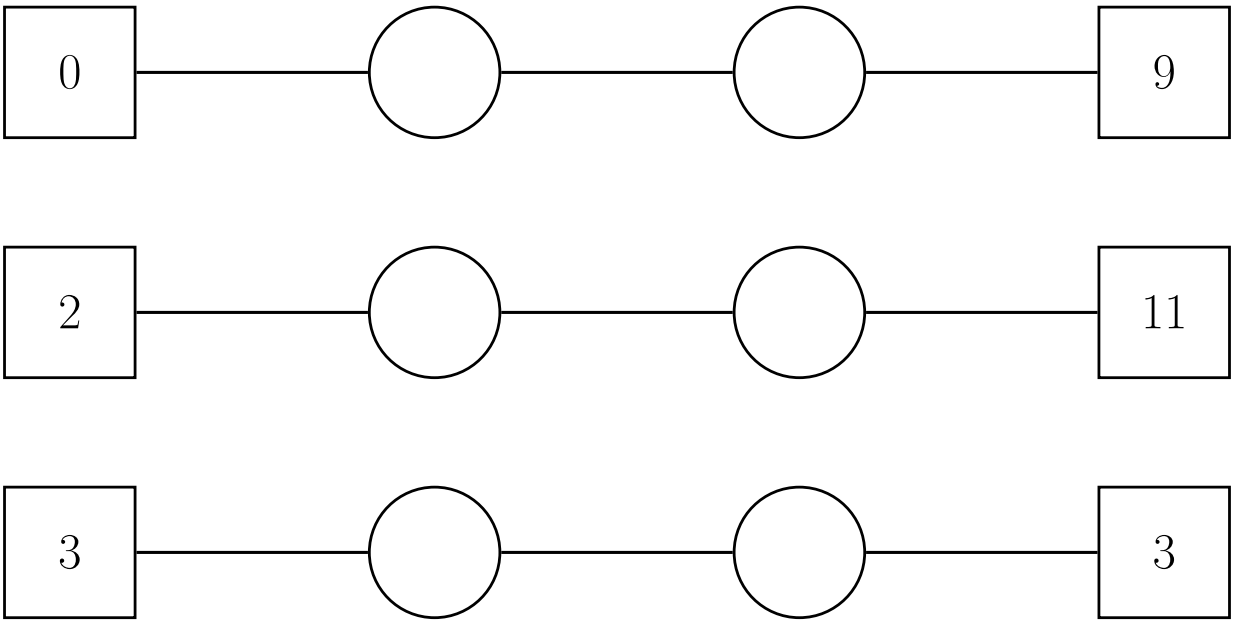


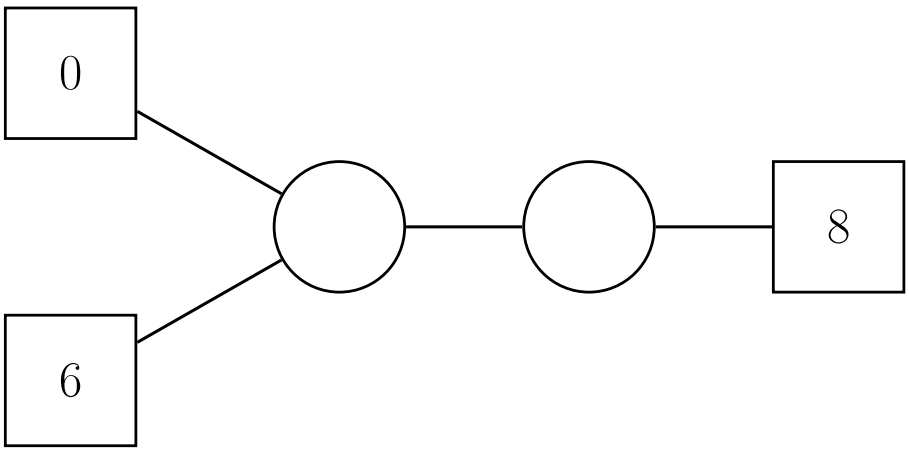


Example: both circles work.

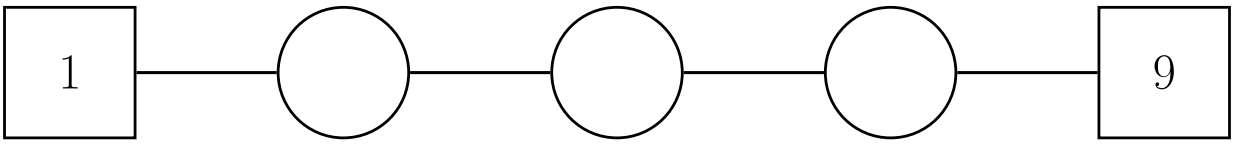


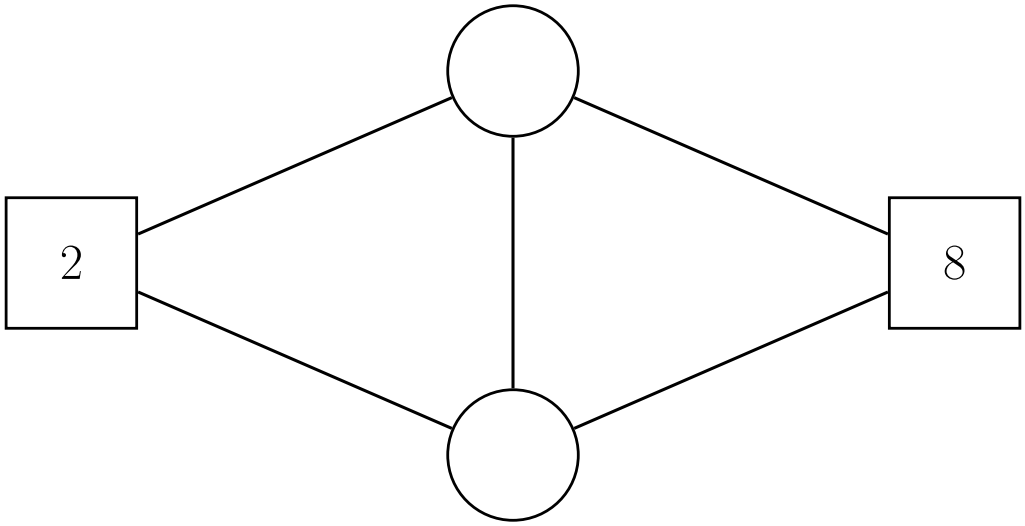
Problem 2: Fill every circle so that all circles on a board work at once.



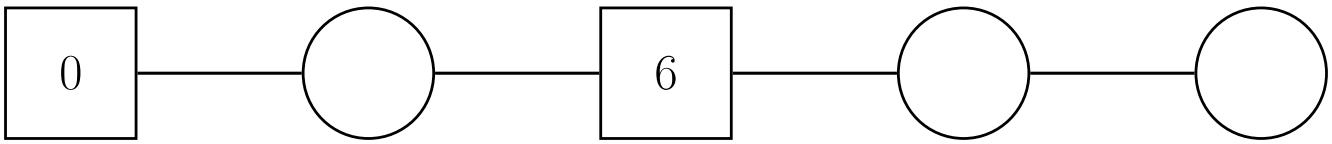


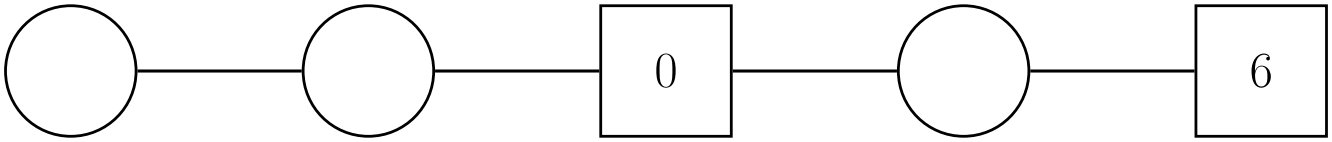
Problem 3: Can you fill either board so that some circle is bigger than every square on its board? Make a filling that works, or explain why there is none.



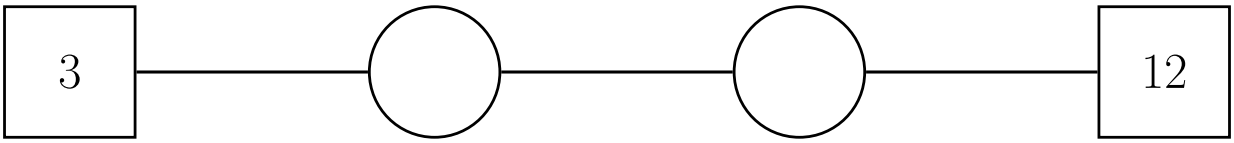


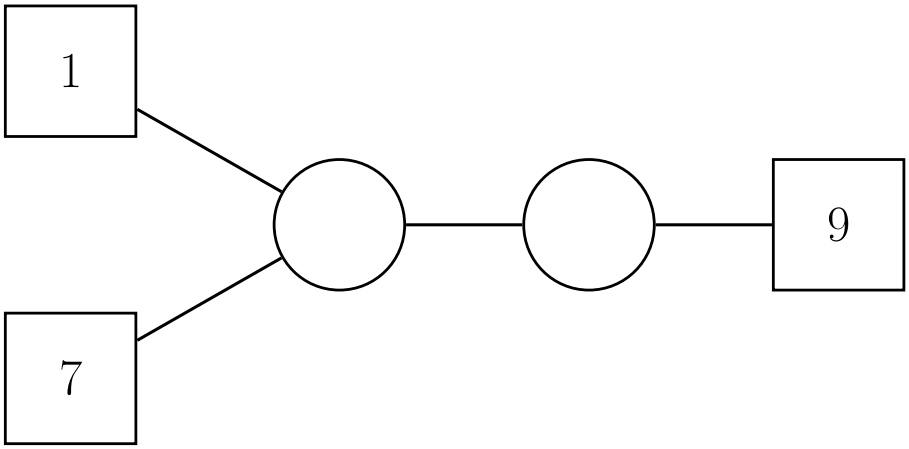
Problem 4: Fill all the circles. Can a circle match the biggest square on its board? Can a circle match the smallest square on its board?

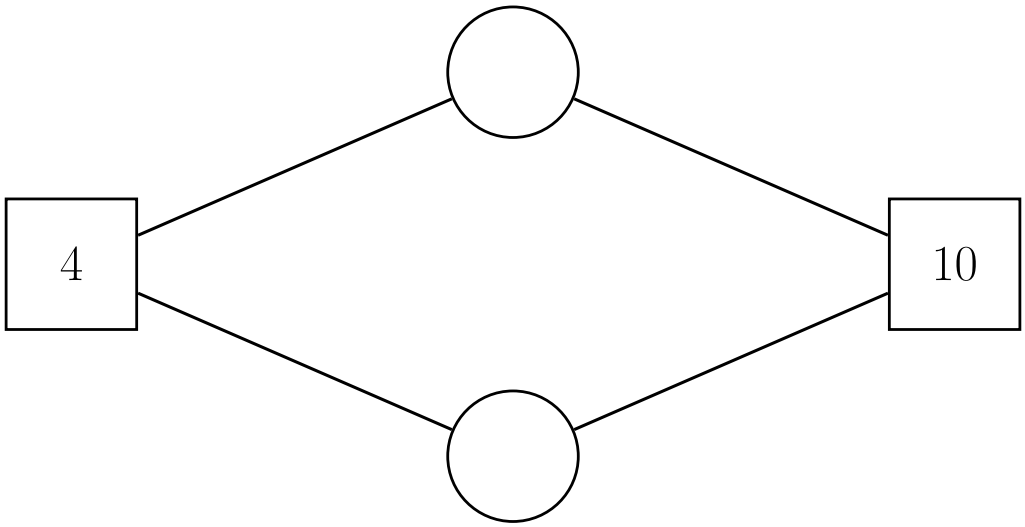




Problem 5: Fill each board. Can any board have a different filling with the same square numbers? Find one, or explain why it cannot.







Problem 6: Use whole numbers from 0 to 5. Find every filling for each board, and explain how you know none are missing.

